

A Complete Overview of All Experiments

Sentence-level word Confidence for GPT-3.5-Turbo and GPT-4-Turbo

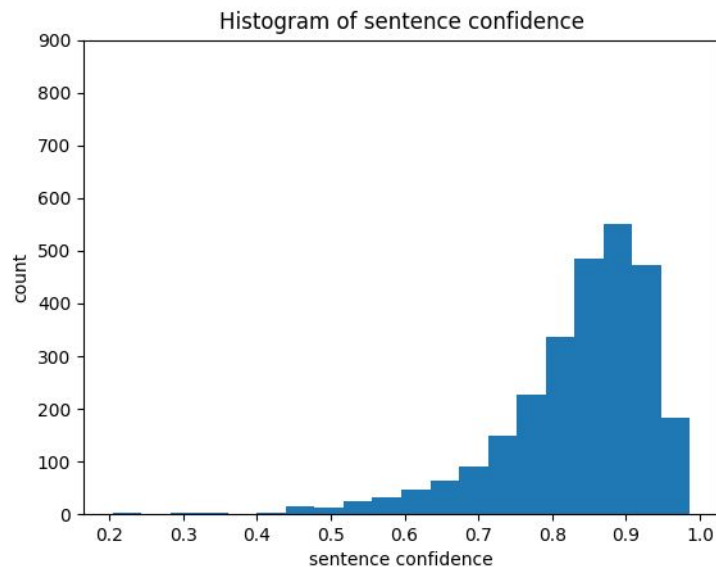
Here is the example of prompt for this experiment

```
{  
    'role': 'system',  
    'content': f""""You are a helpful assistant that corrects ASR errors. \  
You will be presented with an ASR transcription delimited by {delimiter} characters,\  
and your task is to rectify any errors in it. \  
Provide the most probable corrected transcription in string format. \  
Do not change the case, for example, lower case or upper case, in the transcription. \  
Do not output any additional text that is not the corrected transcription. \  
Do not write any explanatory text that is not the corrected transcription.  
""""  
}
```

Here is the one example of our output for first step

```
{  
  "asr_transcription": {  
    "text": " On the I'll do it more completely.",  
    "confidence_score": 0.79  
  },  
  "reference_transcription": "ONLY I'LL DO IT MORE COMPLETELY",  
  "corrected_asr_transcription": "I'll do it more completely."  
}
```

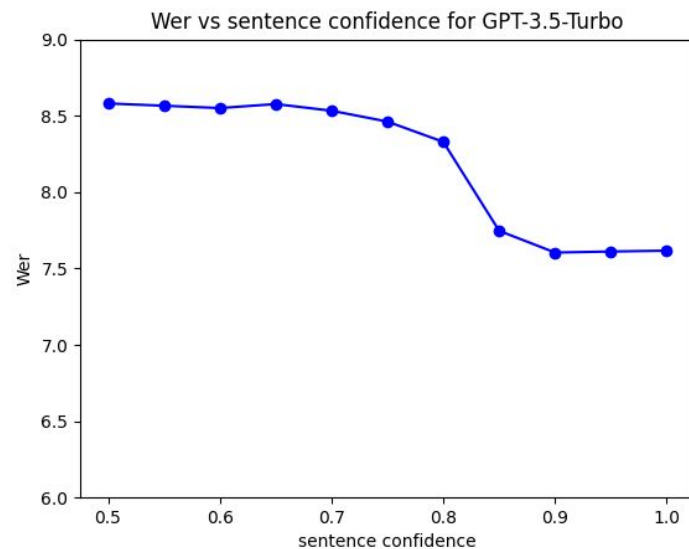
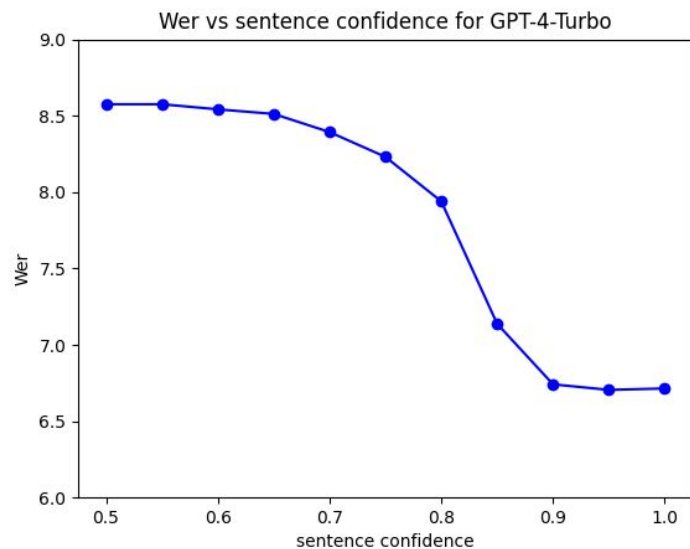
Histogram of the sentence-level confidence scores of the Librispeech transcriptions



Based on the histogram on last slide, we changed the range of threshold.

Threshold = [0.5, 0.55, 0.6, 0.65, ..., 1.0]

Wer vs sentence confidence plots for GPT-3.5-Turbo and GPT-4-Turbo

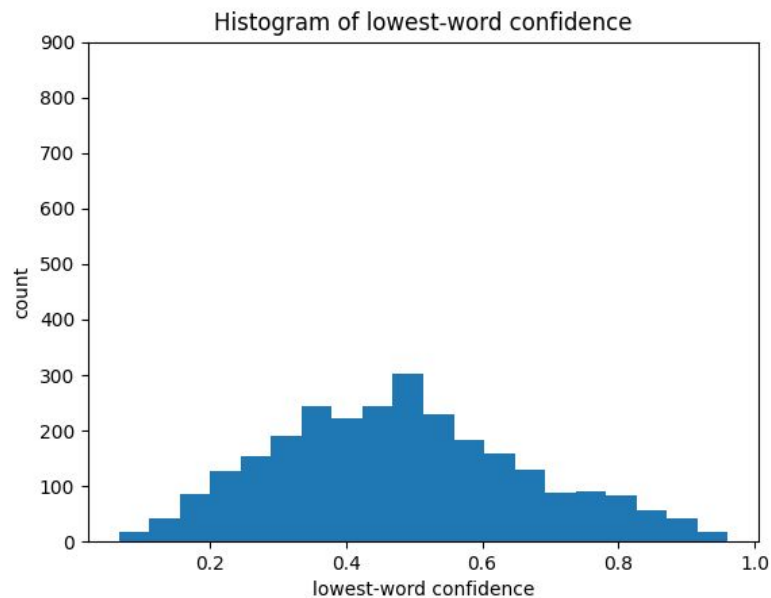


Lowest-Confidence Words in GPT-3.5-Turbo and GPT-4-Turbo

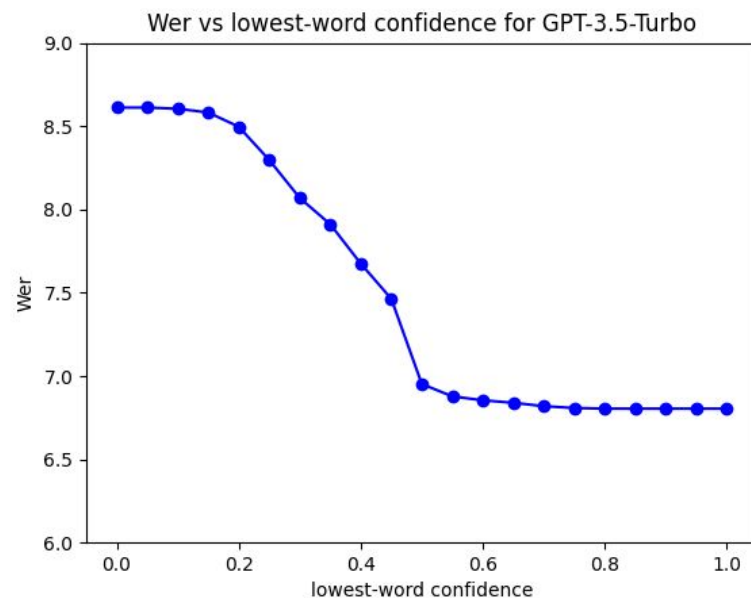
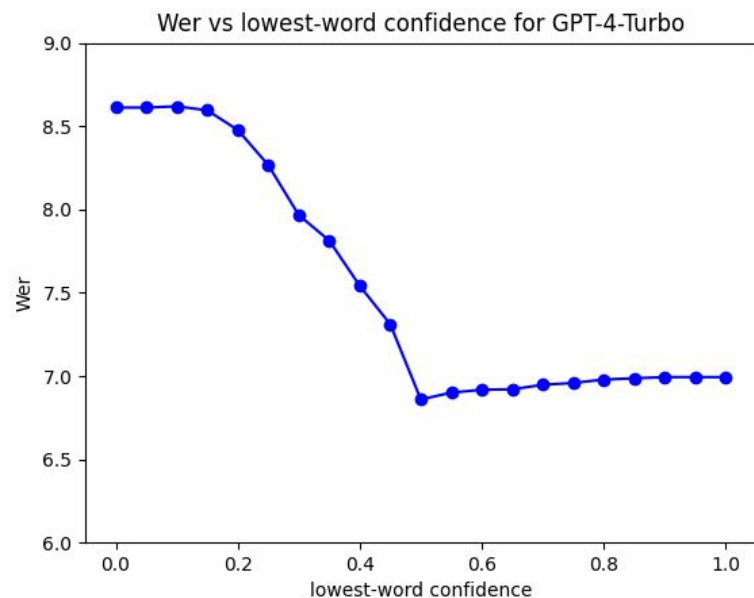
Here is the example of prompt for next experiment

```
{  
    'role': 'system',  
    'content': f""""You are a helpful assistant that corrects ASR errors. \  
You will be presented with an ASR transcription delimited by {delimiter} characters,\  
and your task is to rectify any errors in it. \  
Provide the most probable corrected transcription in string format. \  
Do not change the case, for example, lower case or upper case, in the transcription. \  
Do not output any additional text that is not the corrected transcription. \  
Do not write any explanatory text that is not the corrected transcription.  
""""  
}
```

Histogram plot of the lowest-confidence words of the Librispeech transcriptions



WER vs lowest-confidence plots for GPT-3.5-Turbo and GPT-4-Turbo



Certain Low Confidence for
GPT-3.5-Turbo and GPT-4-Turbo

Here is the prompt of this experiment

```
{  'role': 'system',
    'content': f"""You are a helpful assistant that corrects ASR errors. \
You will be presented with an ASR transcription and low confidence words in that transcription. \
the input will be formatted as json with keys text and low_confidence_words,\
where the text is the ASR transcription and low_confidence_words contains the list of words with low asr confidence. \
your task is to check if the low confidence words make sense in the transcription and if they do not, replace them with better words. \
provide your output as a string. \
Do not change the case, for example, lower case or upper case, in the transcription. \
Do not output any additional text that is not the corrected transcription. \
Do not write any explanatory text that is not the corrected transcription.
    """}
```

Providing example in prompt

```
{  
    'role': 'user',  
    'content': '{"text": "Why not allow your silver tuff to luxuriate in a natural manner?",  
"low_confidence_words":["tuff"]}'  
},  
{'role': 'assistant', 'content': "why not allow your silver tufts to luxuriate in a natural manner?"},  
{'role': 'user', 'content': json.dumps(asr_transcription)}  
}
```

Code snippet

```
for i,d in enumerate(data):
```

```
    asr_transcription = confidence_score_word_level(d["asr_transcription"], confidence = True)
```

```
    low_confidence_words = [word["text"] for word in asr_transcription["words"]
```

```
        if word["confidence"] <= THRESH]
```

```
    if len(low_confidence_words) > 0:
```

```
        asr_transcription = {"text": asr_transcription["text"], "low_confidence_words": low_confidence_words}
```

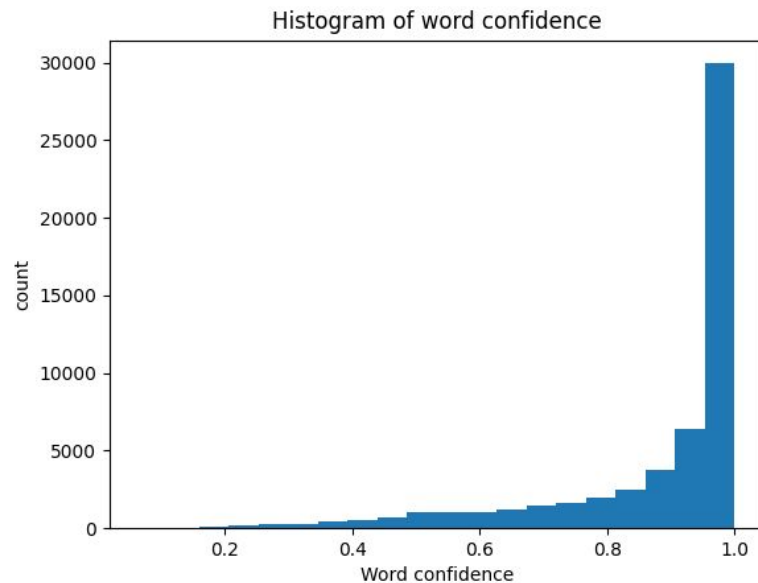
```
        reference_transcription= d["reference_transcription"]
```

```
        data1.append({"asr_transcription": asr_transcription, "reference_transcription": reference_transcription})
```

Here is the one example of output for this step

```
{  
  "asr_transcription": {  
    "text": " But, Dukalion and Piro were very sad. For they knew that they were the only persons who were left alive in all the land.",  
    "low_confidence_words": [  
      "Dukalion",  
      "Piro"  
    ]  
  },  
  "reference_transcription": "BUT DEUCALION AND PYRRHA WERE VERY SAD FOR THEY KNEW THAT THEY WERE THE ONLY PERSONS WHO WERE LEFT  
ALIVE IN ALL THE LAND",  
  "corrected_asr_transcription": "But, Deucalion and Pyrrha were very sad. For they knew that they were the only persons who were left alive in all the land."  
},
```

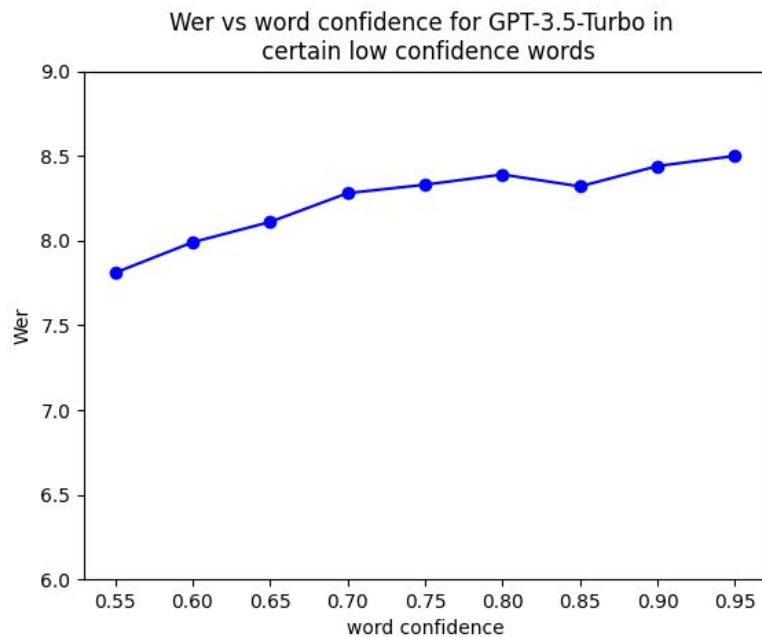

Histogram of the word-level confidence scores of the Librispeech transcriptions



Errors for certain low confidence experiment for GPT-3.5-Turbo

	Tresh	WER_corrected	WER_original	CER_corrected	CER_original	SER_corrected	SER_original
0	0.55	7.81	8.61	3.74	3.57	61.56	61.45
1	0.60	7.99	8.61	3.91	3.57	62.82	61.45
2	0.65	8.11	8.61	4.01	3.57	63.70	61.45
3	0.70	8.28	8.61	4.10	3.57	64.48	61.45
4	0.75	8.33	8.61	4.17	3.57	64.96	61.45
5	0.80	8.39	8.61	4.16	3.57	66.26	61.45
6	0.85	8.32	8.61	4.14	3.57	65.45	61.45
7	0.90	8.44	8.61	4.19	3.57	65.00	61.45
8	0.95	8.50	8.61	4.14	3.57	64.96	61.45

Wer vs certain low confidence plots for GPT-3.5-Turbo



Erros for both GPT-4-Turbo and GPT-3.5-Turbo(Thresh = 0.6)

GPT-4-Turbo

WER_original:	8.6116%
WER_corrected_chatgpt:	6.8652%
CER_original:	3.5707%
CER_corrected_chatgpt:	3.4416%
SER_original:	61.4502%
SER_corrected_chatgpt:	55.6419%

GPT-3.5-Turbo

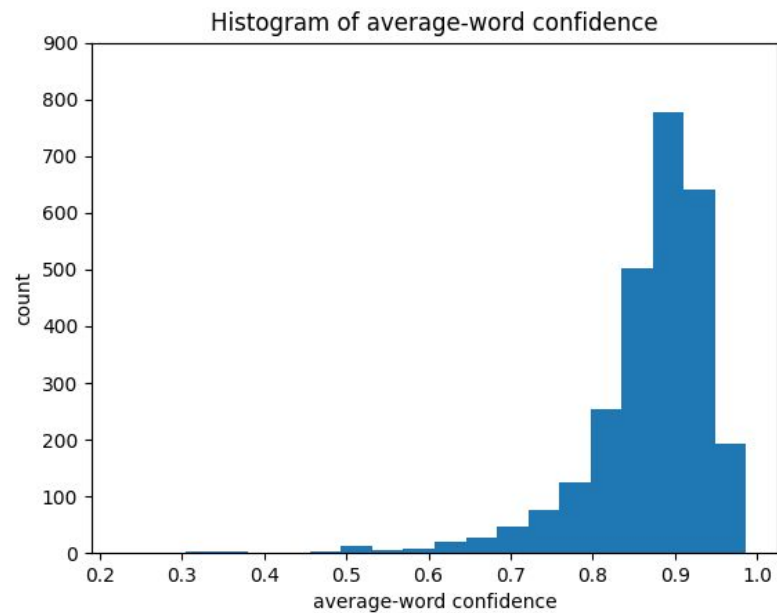
WER_original:	8.6116%
WER_corrected_chatgpt:	7.9853%
CER_original:	3.5707%
CER_corrected_chatgpt:	3.9074%
SER_original:	61.4502%
SER_corrected_chatgpt:	62.8191%

Average-level Confidence for GPT-3.5-Turbo and GPT-4-Turbo

Here is the example of prompt for this experiment

```
{  
  'role': 'system',  
  'content': f""""You are a helpful assistant that corrects ASR errors. \  
You will be presented with an ASR transcription delimited by {delimiter} characters,\  
and your task is to rectify any errors in it. \  
Provide the most probable corrected transcription in string format. \  
Do not change the case, for example, lower case or upper case, in the transcription. \  
Do not output any additional text that is not the corrected transcription. \  
Do not write any explanatory text that is not the corrected transcription.  
""""  
}
```

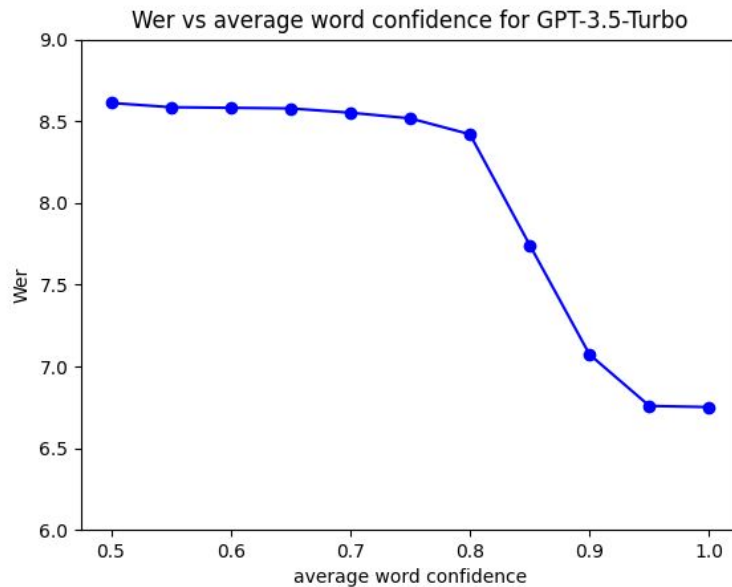
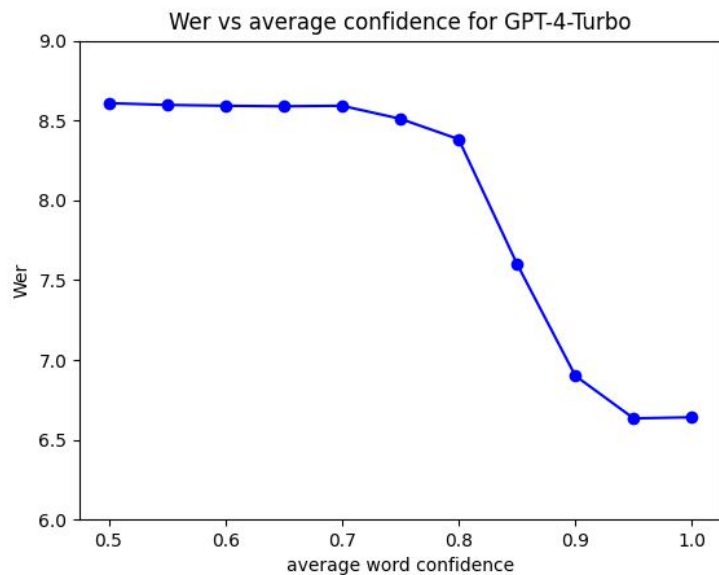
Histogram of the average-level confidence scores of the Librispeech transcriptions



Based on the histogram on last slide, we changed the range of threshold.

Threshold = [0.5, 0.55, 0.6, 0.65, ..., 1.0]

Wer vs average-level confidence plots for GPT-3.5-Turbo and GPT-4-Turbo



My team's suggestions about next step

- They suggested me to try different whisper models (base-small-medium) for both gpt-3.5-turbo-1106 and gpt-4-1106-preview and compare all the results of these model together.
- They also suggested me that we can use LF-MMI or Hybrid-HMM/ANN instead of whisper model.

Thank you!